BIG SCENARIO QUESTION PRACTICE

Cloud Computing & IoT Cloud Technology

40 Important 7-Mark Questions

Concept -> Scenario Evidence -> Theory Application -> Recommendation -> Expected Impact

A	Concept
B	Scenario Evidence
C	Theory Application
D	Recommendation
E	Expected Impact

Exam-ready scenario diagnosis, source-based theory and practical recommendations

Source basis: the uploaded 43-page, 40-question exam handbook and its 103-page lecture PDF. Historical market/vendor statements are retained as lecture content, not current market facts.

How to Answer a Big Scenario Question for 7 Marks

A. Concept (1 mark): Define the exact theory or model in 2-3 lines. Use the lecture terminology and full forms.
B. Scenario Evidence (1 mark): Extract three facts from the case. Do not copy the entire scenario; select only evidence that affects the technical decision.
C. Theory Application (2.5 marks): Match each evidence item to a characteristic, layer, model, actor, workflow step or comparison criterion from the theory.
D. Recommendation (1.5 marks): State what the organization should implement and why. A recommendation must follow from the theory application.
E. Expected Impact (1 mark): Write 2-3 measurable or operational results such as lower latency, reduced cost, improved scalability or clearer control.

Perfect A+ Writing Formula

1. Read the case and underline constraints: cost, latency, security, control, scale, battery, location or responsibility.
2. Name the concept before giving the solution.
3. Use matched wording: Evidence -> Theory -> Decision.
4. Avoid a generic recommendation that is not supported by the scenario.
5. Finish with an impact statement and one-line conclusion.

Mini Example

Scenario: A mobile video application has long delay because all processing occurs in a distant cloud.
A: Edge computing places resources near the base station or access point.
B: Evidence: real-time service, long delay, high network use.
C: Theory: edge is distributed, close to UE, short-latency and lower-jitter, but has limited capacity.
D: Process real-time functions at the edge and retain the central cloud for storage/overflow.
E: Lower latency and jitter, better QoS, reduced battery/network usage.

Exam warning: Do not write only theory. Every theory point must be visibly connected to evidence from the case.

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Each practice uses the same A-E scoring sequence and ends with an examiner-focused result.

Q1 Trace the evolution of cloud computing.

Source basis: uploaded lecture PDF, pages 2-4 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A university still runs one large central server for many departments. Remote campuses need secure access, separate operating environments and the ability to add capacity only during admission periods. Management asks how computing reached the present cloud model and why migration is logical.

A. Concept [1 mark]: Cloud computing evolved gradually from shared mainframes, networking and virtualization into Internet-based, on-demand computing services.

B. Scenario Evidence [1 mark]

- Many departments share one central computer.
- Remote campuses require network-based access.
- Workloads need isolation and temporary capacity.

C. Theory Application [2.5 marks]

1. Time-shared mainframes explain the original shared-computing arrangement.
2. ARPANET, Internet expansion and VPNs explain remote and secure access.
3. IBM virtual-machine concepts explain how one physical server can host isolated environments.
4. Salesforce, AWS, EC2, Google App Engine and Azure represent the move to rentable online services.

D. Recommendation [1.5 marks]: Migrate in stages: virtualize the current server, connect campuses securely, and move seasonal workloads to an on-demand cloud service while retaining critical local systems during transition.

E. Expected Impact [1 mark]

- Less dependence on one physical machine.
- Rapid capacity during admission peaks.
- Lower capital and maintenance burden.

Conclusion: Thus, cloud computing is the result of convergence among networking, virtualization, distributed computing and utility-style service delivery.

A+ EXAM SUGGESTION: Draw a timeline: Mainframe -> ARPANET -> VM -> Internet/VPN -> Salesforce/AWS/EC2 -> App Engine/Azure.

Q2 Explain recent trends in computing: cluster, grid, utility and cloud computing.

Source basis: uploaded lecture PDF, pages 2 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A research consortium has four requirements: tightly coupled processing inside one laboratory, collaboration among different universities, metered temporary storage, and rapid web-service deployment. The committee is confused about cluster, grid, utility and cloud computing.

A. Concept [1 mark]: The lecture presents four stages of computing that progressively increase resource sharing, geographical distribution and service abstraction.

B. Scenario Evidence [1 mark]

- Local nodes are similar and tightly connected.
- Universities use heterogeneous systems across a wide area.
- Some resources must be consumed and billed on demand.

C. Theory Application [2.5 marks]

1. Cluster computing fits homogeneous nodes working together in one local environment.
2. Grid computing fits heterogeneous, geographically distributed nodes performing large tasks.
3. Utility computing fits packaged compute and storage supplied on a metered basis.
4. Cloud computing fits a shared pool with ubiquitous, dynamic and on-demand access.

D. Recommendation [1.5 marks]: Use a cluster for tightly coupled laboratory jobs, a grid for cross-university computation, and cloud/utility services for elastic web hosting and temporary storage.

E. Expected Impact [1 mark]

- Correct technology for each workload.
- Better resource utilization.
- Faster service provisioning with controlled cost.

Conclusion: Cloud computing extends earlier distributed and utility concepts by adding ubiquitous, elastic and on-demand access.

A+ EXAM SUGGESTION: Use the four-stage arrow: Cluster -> Grid -> Utility -> Cloud.

Q3 Define cloud computing and explain the NIST visual model.

Source basis: uploaded lecture PDF, pages 4-5 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A hospital plans an online patient portal. Doctors will connect from laptops and mobiles, usage will rise during emergencies, departments will share infrastructure, and management requires metered billing. The architect must show whether the design satisfies the NIST cloud model.

A. Concept [1 mark]: According to the NIST-based lecture definition, cloud computing enables convenient, on-demand network access to a shared pool of configurable computing resources.

B. Scenario Evidence [1 mark]

- Access is required through heterogeneous networked devices.
- Emergency demand changes rapidly.
- Shared infrastructure and usage measurement are required.

C. Theory Application [2.5 marks]

1. Broad network access supports mobiles, laptops and standard interfaces.
2. Rapid elasticity provides automatic scale-out and scale-in during emergencies.
3. Resource pooling enables multi-tenant sharing across departments.
4. Measured service and on-demand self-service provide transparent usage control and automatic provisioning.
5. The solution can be classified further by service model and deployment model.

D. Recommendation [1.5 marks]: Adopt a cloud design that explicitly implements all five essential characteristics, then document whether the portal is SaaS/PaaS/IaaS and public/private/hybrid/community.

E. Expected Impact [1 mark]

- A defensible NIST-aligned architecture.
- Reliable emergency scaling.
- Transparent consumption and governance.

Conclusion: The NIST visual model describes cloud computing through three dimensions: characteristics, service models and deployment models.

A+ EXAM SUGGESTION: Draw three boxes: Essential Characteristics | Service Models | Deployment Models.

Q4 Discuss the business advantages and general characteristics of cloud computing.

Source basis: uploaded lecture PDF, pages 6-7 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A small retailer wants to launch an online store before a festival. It has little capital, uncertain traffic and no large IT team. The owner asks whether cloud computing offers both business and technical advantages.

A. Concept [1 mark]: Cloud computing provides financial, operational and technical benefits by converting infrastructure into an accessible, shared service.

B. Scenario Evidence [1 mark]

- Upfront budget is limited.
- Demand is uncertain and time-sensitive.
- The company lacks maintenance staff.

C. Theory Application [2.5 marks]

1. Nearly zero upfront infrastructure investment avoids a new data center.
2. Real-time availability and agility shorten provisioning time.
3. Elasticity, load balancing and scalability respond to festival traffic.
4. Usage-based costing and multitenancy reduce idle-resource waste.
5. Low maintenance and ubiquitous access support a small distributed team.

D. Recommendation [1.5 marks]: Launch on a pay-per-use cloud platform with auto-scaling, monitoring and access controls; begin small and increase resources only when measured demand rises.

E. Expected Impact [1 mark]

- Reduced time to market.
- Lower financial risk.
- Improved reliability and resource efficiency.

Conclusion: Cloud computing improves business responsiveness while reducing infrastructure cost, maintenance effort and resource wastage.

A+ EXAM SUGGESTION: Separate the answer into Business Advantages and General Characteristics.

Q5 Explain the five essential characteristics of cloud computing.

Source basis: uploaded lecture PDF, pages 8-12 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: An e-learning provider claims its ordinary hosted server is a cloud. Students can access it online, but capacity changes require a technician, resources are dedicated, and usage is not measured. Evaluate the claim using the five essential characteristics.

A. Concept [1 mark]: The NIST model identifies five characteristics that distinguish a true cloud service from ordinary remote hosting.

B. Scenario Evidence [1 mark]

- Network access exists.
- Provisioning is manual and capacity is fixed.
- No pooling or metering is available.

C. Theory Application [2.5 marks]

1. Broad network access is satisfied because students reach the service online.
2. Rapid elasticity is not satisfied because scale changes are not automatic.
3. Measured service is absent because resource usage is not recorded transparently.
4. On-demand self-service is absent because a technician must provision capacity.
5. Resource pooling is absent because the server is dedicated rather than multi-tenant.

D. Recommendation [1.5 marks]: Add automated self-service provisioning, pooled virtual resources, elastic scaling and metering; otherwise describe the system as remote hosting, not a complete NIST cloud.

E. Expected Impact [1 mark]

- Accurate technology classification.
- Faster response to enrolment peaks.
- Improved utilization and charge transparency.

Conclusion: Together, these five characteristics provide accessibility, automation, elasticity, sharing and accountable consumption.

A+ EXAM SUGGESTION: Use the mnemonic B-R-M-O-R: Broad, Rapid, Measured, On-demand, Resource pooling.

Q6 Explain the components of cloud computing.

Source basis: uploaded lecture PDF, pages 13 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A municipality is building a citizen-service cloud. Citizens use phones, applications need identity and mapping, developers require hosting, records need databases, and the system needs virtualized servers. The team must identify every cloud component.

A. Concept [1 mark]: The cloud computing environment is organized as a layered collection of clients, services, applications, platforms, storage and infrastructure.

B. Scenario Evidence [1 mark]

- Multiple client types access the system.
- Applications depend on platform and data services.
- Virtualized compute, storage and networking are required.

C. Theory Application [2.5 marks]

1. Clients include thick, thin and mobile end-user devices.
2. Services include identity, mapping and search functions.
3. Applications include web apps and SaaS solutions.
4. The platform hosts and deploys applications through PaaS.
5. Storage provides databases and data-storage services.
6. Infrastructure provides virtualization, compute and IaaS resources.

D. Recommendation [1.5 marks]: Document the design as a six-layer stack and assign ownership, interfaces and security controls to every layer before implementation.

E. Expected Impact [1 mark]

- Clear architecture and responsibility.
- Easier troubleshooting and scaling.
- Better integration between citizen apps and infrastructure.

Conclusion: Each component performs a distinct role, but all layers cooperate to deliver cloud services to the end-user.

A+ EXAM SUGGESTION: Draw: Clients -> Services -> Applications -> Platform -> Storage -> Infrastructure.

Q7 Compare SaaS, PaaS and IaaS service models.

Source basis: uploaded lecture PDF, pages 14-17, 28-30, 49 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A company needs three solutions: employees want ready-to-use email, developers want a managed environment to deploy code, and administrators want virtual machines with operating-system control. Select the correct service model for each.

A. Concept [1 mark]: Cloud service models differ mainly in the level of abstraction provided and the amount of control retained by the user.

B. Scenario Evidence [1 mark]

- Employees need a complete application.
- Developers need a platform, not server administration.
- Administrators require OS and network-level control.

C. Theory Application [2.5 marks]

1. SaaS delivers the complete email application and provider-managed stack.
2. PaaS supplies runtime, middleware, deployment tools and infrastructure while developers control applications and data.
3. IaaS supplies networking, storage, servers and virtualization while the user manages OS, middleware and applications.
4. Control increases from SaaS to PaaS to IaaS, while provider responsibility decreases.

D. Recommendation [1.5 marks]: Use SaaS for email, PaaS for application development and IaaS for custom server workloads that genuinely need administrative control.

E. Expected Impact [1 mark]

- No unnecessary management overhead.
- Faster development and deployment.
- Appropriate control for specialized workloads.

Conclusion: SaaS, PaaS and IaaS form a service stack from complete applications down to basic infrastructure.

A+ EXAM SUGGESTION: Compare provides, user control, provider control and example.

Q8 Explain the architecture of cloud service models and their self-service interfaces.

Source basis: uploaded lecture PDF, pages 28 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A cloud provider has applications at the top, platform tools in the middle, virtual machines and storage at the bottom, and one administration service for packaging, deployment, scaling and users. Customers complain that they cannot request resources directly.

A. Concept [1 mark]: The lecture architecture shows SaaS, PaaS and IaaS as connected layers supported by self-service interfaces and a common administration service.

B. Scenario Evidence [1 mark]

- The service layers already exist.
- Administration functions are centralized.
- Self-service access is missing.

C. Theory Application [2.5 marks]

1. SaaS should expose complete applications to clients.
2. The PaaS self-service interface should expose security/identity, integration workflow, UI services, database grid and application grid.
3. The IaaS self-service interface should expose virtual machines, virtual storage and virtual grids.
4. The administration service should handle packaging, configuration, deployment, scaling, lifecycle, utilization and user management.

D. Recommendation [1.5 marks]: Add role-based PaaS and IaaS self-service portals connected to the administration service, with quotas and metering for controlled automation.

E. Expected Impact [1 mark]

- Reduced manual provisioning delay.
- Consistent lifecycle management.
- Better customer control without exposing physical infrastructure.

Conclusion: The architecture integrates layered services, self-service resource access and centralized administration.

A+ EXAM SUGGESTION: Draw SaaS/PaaS/IaaS in the center with PaaS and IaaS self-service interfaces.

Q9 Explain Software-as-a-Service (SaaS).

Source basis: uploaded lecture PDF, pages 15, 38, 55-57 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A college wants all students to use the same office application through browsers. It cannot install software on every device and wants the provider to handle updates, maintenance and scaling.

A. Concept [1 mark]: SaaS is a cloud model in which a provider hosts a software application and makes it available to users through the Internet.

B. Scenario Evidence [1 mark]

- A complete application is required.
- Access must work through browsers and different devices.
- Local installation and maintenance should be avoided.

C. Theory Application [2.5 marks]

1. SaaS delivers the provider's application as an online service.
2. Users access it through browser or app clients.
3. The provider controls the cloud infrastructure, updates and operational maintenance.
4. On-demand use, centralized data and multitenancy support many students.
5. Examples in the lecture include Google Apps, Salesforce, Learn.com and Office 365.

D. Recommendation [1.5 marks]: Adopt a SaaS solution with institution-managed identities, access policies, backup rules and a clear subscription plan.

E. Expected Impact [1 mark]

- Uniform software for all students.
- Lower installation and support effort.
- Quick scaling for enrolment changes.

Conclusion: SaaS offers convenient, centrally managed software with minimal technical responsibility for the end-user.

A+ EXAM SUGGESTION: Underline browser access, provider-managed, multitenancy, on-demand and pay-per-use.

Q10 Explain Platform-as-a-Service (PaaS), its types and benefits.

Source basis: uploaded lecture PDF, pages 16, 38, 52-55 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A startup writes web applications but spends too much time configuring operating systems, middleware and deployment servers. It needs automatic scaling and an environment where developers can focus on code.

A. Concept [1 mark]: PaaS provides a managed environment in which consumers can develop, deploy and run their own or acquired applications.

B. Scenario Evidence [1 mark]

- Developers create and deploy their own applications.
- Infrastructure administration causes delay.
- Automatic scaling and security are required.

C. Theory Application [2.5 marks]

1. PaaS provides development and deployment tools plus a managed runtime.
2. The user controls applications and configuration but not the underlying infrastructure.
3. An application delivery-only environment supports on-demand scaling and application security.
4. A stand-alone environment provides a platform for a specific function.
5. Open PaaS and add-on development facilities extend platform flexibility.

D. Recommendation [1.5 marks]: Use PaaS for standard web applications, retain source and data portability, and use IaaS only for components requiring unsupported OS or middleware control.

E. Expected Impact [1 mark]

- Shorter development cycles.
- Less infrastructure administration.
- Reliable deployment and scaling.

Conclusion: PaaS accelerates software development by supplying a scalable, maintained platform while hiding infrastructure complexity.

A+ EXAM SUGGESTION: Include definition, PaaS types, benefits and an example such as Azure or Google App Engine.

Q11 Explain Infrastructure-as-a-Service (IaaS) and its working methodology.

Source basis: uploaded lecture PDF, pages 17, 30-31, 50-52 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A software company must run a legacy operating system, install custom middleware, configure host firewalls and expand storage temporarily. A complete SaaS or managed PaaS will not provide enough control.

A. Concept [1 mark]: IaaS is the on-demand delivery of basic computing infrastructure through the Internet.

B. Scenario Evidence [1 mark]

- Operating-system administration is required.
- Custom software and firewall control are needed.
- Compute and storage demand changes over time.

C. Theory Application [2.5 marks]

1. IaaS provides compute, storage, network and operating-system support on demand.
2. Administrative access to VMs permits installation and control of software.
3. Virtualization places multiple VMs above a hypervisor on shared hardware.
4. The provider manages physical resources while the user manages OS, middleware, data and applications.
5. Resources are rented as an outsourced, on-demand service.

D. Recommendation [1.5 marks]: Select IaaS VMs with the required legacy OS, attach elastic storage, define firewall rules and automate backup, patching and monitoring.

E. Expected Impact [1 mark]

- Required technical control is preserved.
- Hardware purchase is avoided.
- Capacity can be changed without rebuilding the system.

Conclusion: IaaS gives users high flexibility and administrative control without requiring ownership of physical infrastructure.

A+ EXAM SUGGESTION: Draw Physical Server vs Hardware -> Hypervisor -> VM1/VM2.

Q12 Explain cloud deployment models.

Source basis: uploaded lecture PDF, pages 18 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A national education program has public learning materials, confidential examination data, shared research resources among universities and temporary workloads that may move between environments. Choose suitable deployment models.

A. Concept [1 mark]: Deployment models specify who owns, manages and accesses a cloud environment and where its infrastructure is located.

B. Scenario Evidence [1 mark]

- Public content has broad access.
- Examination data requires dedicated control.
- Universities share concerns and some workloads need portability.

C. Theory Application [2.5 marks]

1. Public cloud suits general learning services offered to many users.
2. Private cloud suits confidential examination systems for one organization.
3. Community cloud suits universities with shared security, compliance or jurisdiction concerns.
4. Hybrid cloud joins distinct clouds and supports data/application portability.
5. Distributed, multi-cloud and inter-cloud models address scattered resources or multiple providers.

D. Recommendation [1.5 marks]: Use a hybrid design: public cloud for open content, private cloud for examinations, and a community segment for shared research; define controlled transfer between them.

E. Expected Impact [1 mark]

- Security aligned with data sensitivity.
- Scalable public access.
- Cross-institution collaboration with portability.

Conclusion: The correct deployment model depends on access, control, security, cost and interoperability requirements.

A+ EXAM SUGGESTION: Start with public, private, hybrid and community, then mention other models.

Q13 Explain public cloud and private cloud, and compare them.

Source basis: uploaded lecture PDF, pages 19-21 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A bank and a public news website are selecting cloud environments. The bank values control and security, while the news site values low setup cost and rapid scaling for unpredictable traffic.

A. Concept [1 mark]: Public and private clouds differ mainly in ownership, sharing, customer scope, connectivity and security control.

B. Scenario Evidence [1 mark]

- The bank serves a limited trusted user group.
- The news site serves many public users.
- Security and scalability priorities differ.

C. Theory Application [2.5 marks]

1. Private cloud is dedicated to one organization and can use an internal/private network.
2. Private cloud provides greater control and fewer sharing concerns but requires regular maintenance.
3. Public cloud shares virtualized resources among multiple customers over the Internet.
4. Public cloud offers low setup cost, scalability and pay-per-use, but less dedicated control.

D. Recommendation [1.5 marks]: Use private cloud for core banking workloads and public cloud for the news site; use strong governance and consider hybrid integration for non-sensitive bank services.

E. Expected Impact [1 mark]

- Bank data receives dedicated control.
- The news site scales economically.
- Each organization avoids a one-model-fits-all decision.

Conclusion: Public cloud favors cost and scalability, whereas private cloud favors dedicated control and security.

A+ EXAM SUGGESTION: Use a two-column comparison: resources, customers, connectivity, security, advantage and limitation.

Q14 Explain hybrid, community, distributed, multi-cloud and inter-cloud.

Source basis: uploaded lecture PDF, pages 22-24 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A disaster-management alliance includes government agencies, hospitals and telecom operators. Sensors are scattered nationwide, services must survive provider failure, and applications must move between private and public resources.

A. Concept [1 mark]: Beyond basic public and private clouds, several models combine or distribute resources to meet specialized needs.

B. Scenario Evidence [1 mark]

- Organizations share common emergency concerns.
- Computing devices are geographically scattered.
- Portability and provider fault tolerance are required.

C. Theory Application [2.5 marks]

1. Community cloud supports organizations with common security and compliance concerns.
2. Distributed cloud connects scattered devices through one network.
3. Hybrid cloud joins unique cloud setups and supports application/data portability.
4. Multi-cloud uses multiple provider services to improve flexibility and fault tolerance.
5. Inter-cloud represents a global cloud-of-clouds with provider interoperability.

D. Recommendation [1.5 marks]: Build a community-governed multi-cloud solution, connect field resources as a distributed cloud, and use hybrid portability rules between protected and public services.

E. Expected Impact [1 mark]

- Greater resilience during disasters.
- Shared cost and governance.
- Flexible placement of data and applications.

Conclusion: These models extend cloud deployment through sharing, distribution, integration and provider interoperability.

A+ EXAM SUGGESTION: Use the keywords: portability, common concerns, scattered devices, multiple providers, cloud-of-clouds.

Q15 Compare different cloud deployment models using location and access.

Source basis: uploaded lecture PDF, pages 25 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: Four departments propose cloud options: a dedicated system in their own building, a dedicated system hosted by a vendor, a jointly shared system in their own premises, and a broadly shared system hosted externally. Classify each using location and access.

A. Concept [1 mark]: The source classifies deployment models using two dimensions: infrastructure location and whether access is dedicated or shared.

B. Scenario Evidence [1 mark]

- Location is either on-premise or off-premise.
- Access is either dedicated or shared.
- The four combinations must map to deployment models.

C. Theory Application [2.5 marks]

1. On-premise plus dedicated access is private cloud.
2. Off-premise plus dedicated access is hosted private cloud.
3. On-premise plus shared access is community cloud.
4. Off-premise plus shared access is public cloud.

D. Recommendation [1.5 marks]: Use the 2 x 2 location-access matrix as the primary classification tool, then verify ownership, governance and security before final selection.

E. Expected Impact [1 mark]

- Clear classification without terminology confusion.
- Better alignment of access and hosting.
- Faster architectural decision-making.

Conclusion: The location-access matrix provides a simple method for distinguishing private, hosted private, community and public clouds.

A+ EXAM SUGGESTION: Reproduce the exact 2 x 2 matrix in the exam.

Q16 Compare the responsibility of user and service provider in traditional IT, IaaS, PaaS and SaaS.

Source basis: uploaded lecture PDF, pages 29 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: An enterprise is moving a business application from its own data center to cloud services. The CIO wants to know which layers remain under user control in traditional IT, IaaS, PaaS and SaaS.

A. Concept [1 mark]: As the service model moves from traditional IT to SaaS, more system layers are managed by the cloud service provider.

B. Scenario Evidence [1 mark]

- The current team manages the entire stack.
- Management wants to reduce operational responsibility.
- Some applications may still need control over OS or code.

C. Theory Application [2.5 marks]

1. Traditional IT leaves applications, data, runtime, middleware, OS, virtualization, servers, storage and networking with the user.
2. IaaS moves networking, storage, servers and virtualization to the provider.
3. PaaS also moves OS, middleware and runtime to the provider; the user keeps applications and data.
4. SaaS places the entire stack under provider management, with the user mainly consuming/configuring the service.

D. Recommendation [1.5 marks]: Choose the highest service model that still preserves required control: IaaS for OS-dependent workloads, PaaS for custom applications, and SaaS for standard business functions.

E. Expected Impact [1 mark]

- Reduced administration.
- Clear accountability for each layer.
- Lower risk of selecting an over-managed or under-controlled model.

Conclusion: Responsibility progressively shifts from the user to the provider in the order Traditional -> IaaS -> PaaS -> SaaS.

A+ EXAM SUGGESTION: Draw a nine-layer stack and mark user-managed versus provider-managed.

Q17 Discuss IaaS procedures and benefits.

Source basis: uploaded lecture PDF, pages 50-52 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A seasonal analytics team needs virtual desktops, storage, bandwidth, IP addresses, firewalls and monitoring for three months. Team members must install software and start web servers themselves.

A. Concept [1 mark]: IaaS enables users to obtain virtualized infrastructure, administer it and pay according to the rented resources.

B. Scenario Evidence [1 mark]

- Infrastructure is temporary and rentable.
- Users need VM administrative commands.
- Billing should depend on resource and usage.

C. Theory Application [2.5 marks]

1. IaaS allocates a VM and related resources after a user request.
2. Administrative access supports saving data, starting servers and installing applications.
3. Rentable resources include VMs, storage, bandwidth, firewalls, IP addresses and monitoring.
4. Payment may depend on resource type, usage time and service package.
5. Virtualization provides portability and interoperability with legacy applications.

D. Recommendation [1.5 marks]: Provision standardized VM images, role-based administration, usage monitoring and automatic shutdown at project completion.

E. Expected Impact [1 mark]

- Fast project startup.
- Only three months of infrastructure cost.
- Controlled administration and easier resource release.

Conclusion: IaaS combines administrative flexibility, resource rental, portability and compatibility with existing workloads.

A+ EXAM SUGGESTION: Mention three VM commands and at least four rentable resources.

Q18 Explain SaaS characteristics and advantages.

Source basis: uploaded lecture PDF, pages 55-57 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A distributed sales company wants one CRM system for employees in many locations. It expects frequent updates, fluctuating user numbers and a need to share the same customer data securely.

A. Concept [1 mark]: SaaS delivers complete software through the cloud and centralizes operation, maintenance and data management.

B. Scenario Evidence [1 mark]

- Users are geographically distributed.
- Software updates must be centralized.
- The same data and application serve many users.

C. Theory Application [2.5 marks]

1. SaaS delivers software through the Internet from a cloud server.
2. On-demand availability supports access from different locations.
3. The provider performs maintenance, updates and scaling.
4. Centralized data management supports common customer information.
5. Multitenancy and subscription/pay-per-use reduce per-user infrastructure cost.

D. Recommendation [1.5 marks]: Adopt SaaS CRM with identity integration, role-based access, data export arrangements and service availability commitments.

E. Expected Impact [1 mark]

- Consistent CRM access everywhere.
- Lower update and maintenance effort.
- Scalable user onboarding.

Conclusion: SaaS reduces installation and maintenance effort while providing scalable, shared and economical application access.

A+ EXAM SUGGESTION: Explain on-demand, maintenance, scaling, shared data, cost and multitenancy.

Q19 Explain virtualization and its benefits in cloud computing.

Source basis: uploaded lecture PDF, pages 31, 39 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A data center has powerful servers running at low utilization. Different departments require isolated operating systems, and one workload must move to another machine during maintenance.

A. Concept [1 mark]: Virtualization makes one physical computer appear as multiple independent computers called virtual machines.

B. Scenario Evidence [1 mark]

- Physical capacity is underused.
- Departments need isolation.
- Workloads must be portable.

C. Theory Application [2.5 marks]

1. A hypervisor divides physical hardware into multiple virtual machines.
2. Resource sharing improves utilization and reduces cost.
3. Encapsulation packages a complete computing environment.
4. Hardware independence lets VMs run without being tied to one physical machine.
5. VM migration provides portability during maintenance.

D. Recommendation [1.5 marks]: Consolidate workloads on a virtualized cluster, define resource limits and isolation, and enable planned VM migration with tested backups.

E. Expected Impact [1 mark]

- Higher server utilization.
- Reduced hardware cost.
- Maintenance with less service interruption.

Conclusion: Virtualization is a core cloud-enabling technology because it provides isolation, flexibility, portability and efficient sharing.

A+ EXAM SUGGESTION: Underline sharing, encapsulation, independence and portability.

Q20 Explain Wireless Sensor Networks (WSNs), sensor-node components and applications.

Source basis: uploaded lecture PDF, pages 34-35 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A smart-farm project must measure soil moisture across a large field and send readings wirelessly to a control station. Some nodes cannot reach the station directly, and the data will guide irrigation.

A. Concept [1 mark]: A WSN contains sensor nodes that observe physical phenomena and communicate sensed data to a centralized sink node.

B. Scenario Evidence [1 mark]

- Physical phenomena must be sensed.
- Wireless multi-hop communication may be required.
- A central point must aggregate/process data.

C. Theory Application [2.5 marks]

1. The sensing unit measures soil moisture.
2. The processing unit performs local handling of readings.
3. The communication unit transmits data wirelessly.
4. Single-hop or multi-hop paths carry data to the sink node.
5. The sink further processes data for the agriculture application.

D. Recommendation [1.5 marks]: Deploy sensor nodes with suitable sensing, processing and communication units, plan multi-hop routing to a reliable sink, and design energy-aware maintenance.

E. Expected Impact [1 mark]

- Continuous field monitoring.
- Data-driven irrigation.
- Coverage beyond direct radio range.

Conclusion: A WSN combines distributed sensing, local processing and wireless communication to support monitoring applications.

A+ EXAM SUGGESTION: Draw sensor nodes connected to one sink and label the three sensor-node units.

Q21 Discuss the limitations of traditional WSNs.

Source basis: uploaded lecture PDF, pages 40 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A city bought traffic sensors from one vendor, deployed them in fixed locations and now wants to reuse the network for air-quality monitoring. Batteries are failing and changing sensor types would require major redesign.

A. Concept [1 mark]: Traditional WSN deployment creates procurement, deployment, maintenance and requirement-change difficulties for users.

B. Scenario Evidence [1 mark]

- Procurement is vendor- and sensor-type dependent.
- Deployment location/topology is fixed.
- Maintenance and changed requirements are costly.

C. Theory Application [2.5 marks]

1. Procurement limitations include price, vendor selection and integrated sensor types.
2. Deployment limitations include choosing the correct method and place.
3. Post-deployment maintenance and battery lifetime create recurring burden.
4. A change from one application to another can require new sensors, deployment area and topology.

D. Recommendation [1.5 marks]: Before expansion, assess reusable infrastructure, replace only necessary sensors, standardize interfaces and consider a sensor-cloud model for shared access and future flexibility.

E. Expected Impact [1 mark]

- Lower redesign cost.
- Reduced vendor lock-in.
- More adaptable sensing services.

Conclusion: WSN ownership places technical and financial responsibility on the user and is difficult to adapt when requirements change.

A+ EXAM SUGGESTION: Organize under Procurement, Deployment, Maintenance and Change of Requirement.

Q22 Define sensor-cloud and explain its main features.

Source basis: uploaded lecture PDF, pages 33, 36-41 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: Several departments want to use the same environmental sensor network for agriculture, safety and research. They should pay according to use and should not need to know which physical sensor serves each request.

A. Concept [1 mark]: Sensor-cloud is more than merely connecting sensors to a cloud or dumping sensor data into cloud storage; it virtualizes sensor resources as services.

B. Scenario Evidence [1 mark]

- One physical network must serve multiple applications.
- Users need abstraction from physical sensors.
- Usage-based service delivery is required.

C. Theory Application [2.5 marks]

1. Sensor-cloud virtualizes physical sensor nodes and networks.
2. One sensor/WSN can appear as many virtual resources.
3. A service stratum separates sensor owners from end-users.
4. Sensor-as-a-Service provides on-demand access.
5. Pay-per-use supports business sharing rather than simple data dumping to the cloud.

D. Recommendation [1.5 marks]: Introduce virtual sensor instances, a service provider layer, resource scheduling and metering; expose sensing through application templates rather than raw device ownership.

E. Expected Impact [1 mark]

- Shared use of expensive sensors.
- Lower entry cost for applications.
- Flexible and scalable sensing access.

Conclusion: Sensor-cloud converts dedicated sensor networks into virtualized, shareable and service-oriented sensing infrastructure.

A+ EXAM SUGGESTION: Begin by stating that sensor-cloud is not merely dumping sensor data into cloud storage.

Q23 Differentiate WSN and sensor-cloud.

Source basis: uploaded lecture PDF, pages 42-43 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A factory currently owns a WSN dedicated to one production line. Management wants multiple departments to consume the same sensors while another party manages redeployment and maintenance.

A. Concept [1 mark]: A traditional WSN is normally dedicated to one user, while sensor-cloud virtualizes physical sensors for multiple applications and users.

B. Scenario Evidence [1 mark]

- The present network has one dedicated user.
- Multiple applications need the same physical resources.
- Operational responsibility should shift away from the user.

C. Theory Application [2.5 marks]

1. Traditional WSN usage is dedicated to a single WSN user.
2. Sensor-cloud uses virtualization to serve multiple applications/end-users.
3. In WSN, the user handles ownership, deployment, redeployment, maintenance and overhead.
4. In sensor-cloud, the sensor owner owns/deploys devices while the SCSP handles redeployment, maintenance and overhead.
5. End-users consume the service without physical-sensor knowledge.

D. Recommendation [1.5 marks]: Convert the dedicated WSN into a sensor-cloud service with virtual instances, SCSP management and clear owner/user contracts.

E. Expected Impact [1 mark]

- Higher sensor utilization.
- Reduced maintenance burden on departments.
- Support for multiple factory applications.

Conclusion: Sensor-cloud separates ownership, management and usage, reducing user overhead and increasing sharing.

A+ EXAM SUGGESTION: Use a two-column comparison and include ownership, deployment, redeployment, maintenance, overhead and usage.

Q24 Explain the actors and their roles in sensor-cloud.

Source basis: uploaded lecture PDF, pages 43-45 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A sensor-cloud project has three parties: a research group requesting data, a company that buys and deploys sensors, and a platform company that schedules sensors and charges for usage. Define each actor's role.

A. Concept [1 mark]: Sensor-cloud operation depends on three main business and technical actors: end-users, sensor-owners and the Sensor-Cloud Service Provider.

B. Scenario Evidence [1 mark]

- One party consumes applications.
- One party owns physical devices.
- One party manages service delivery and pricing.

C. Theory Application [2.5 marks]

1. End-users consume Sensor-as-a-Service according to application requirements and remain unaware of physical allocation.
2. Sensor-owners purchase and deploy homogeneous or heterogeneous sensors across locations and lend them to the cloud.
3. The Sensor-Cloud Service Provider is a business/managerial actor that allocates resources, operates the platform and charges by usage.
4. The separation of roles enables technical abstraction and a viable business model.

D. Recommendation [1.5 marks]: Write formal service agreements defining sensor ownership, SCSP maintenance/scheduling duties, end-user access rights, pricing and data policies.

E. Expected Impact [1 mark]

- Clear accountability.
- Transparent charging.
- Efficient sensor sharing without ownership confusion.

Conclusion: The actor separation enables users to consume sensing services without owning or directly maintaining the sensor network.

A+ EXAM SUGGESTION: Write each actor under a separate subheading with one business and one technical role.

Q25 Explain the sensor-cloud architecture and views.

Source basis: uploaded lecture PDF, pages 45-46 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A user organization selects a flood-monitoring template in a browser. The request goes through a web portal, physical sensors are selected dynamically, data is aggregated in real time and the resulting feed reaches two applications.

A. Concept [1 mark]: Sensor-cloud architecture connects user organizations, a browser/web portal, cloud management functions and a heterogeneous pool of physical sensors.

B. Scenario Evidence [1 mark]

- Users interact through templates and a browser.
- Physical sensors are selected on demand.
- Aggregated data feeds multiple applications.

C. Theory Application [2.5 marks]

1. The user organization view contains applications, browser interface, login and data feeds.
2. Template display/specification defines the requested sensing service.
3. The web portal interprets requests and receives sensed data.
4. Cloud functions provide dynamic scaling, on-demand scheduling, storage, specialized processing, energy management and QoS.
5. The real view contains a heterogeneous pool of physical sensors.

D. Recommendation [1.5 marks]: Implement clear template definitions, authentication, a web portal, dynamic sensor scheduling and separate real-time aggregation pipelines for each application.

E. Expected Impact [1 mark]

- Simple user interaction.
- Efficient physical-resource allocation.
- Reusable data feeds with controlled QoS.

Conclusion: The architecture hides physical sensor complexity while providing scalable, application-specific sensing services.

A+ EXAM SUGGESTION: Draw User/Application <-> Browser <-> Web Portal/Cloud <-> Physical Sensor Pool.

Q26 Explain the workflow of a sensor-cloud system.

Source basis: uploaded lecture PDF, pages 47 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A user submits an XML template requesting temperature data for two hours. The platform must interpret the request, create a virtual sensor, allocate compatible physical sensors, aggregate data, return it and release resources afterward.

A. Concept [1 mark]: The sensor-cloud workflow creates a virtual sensor from a user request, allocates physical resources, provides data and releases resources when finished.

B. Scenario Evidence [1 mark]

- The request is template-based.
- A virtual sensor must be created from physical resources.
- Allocation and release are both required.

C. Theory Application [2.5 marks]

1. The SensorML interpreter decodes XML definitions, client information and metadata.
2. The Virtual Sensor Manager creates the virtual sensor instance.
3. The Virtual Sensor Controller manages operations and data retrieval.
4. The Resource Manager schedules, allocates and deallocates compatible sensors from the WSN pool.
5. Data is aggregated and provisioned, then the instance and resources are released.

D. Recommendation [1.5 marks]: Automate the full lifecycle with validation, allocation checks, response tracking, timed deletion and resource-release confirmation.

E. Expected Impact [1 mark]

- Correct on-demand sensing.
- No stranded sensor resources.
- Repeatable service delivery from request to deletion.

Conclusion: The workflow provides complete lifecycle management from request and allocation to data delivery and resource release.

A+ EXAM SUGGESTION: Use arrows: User -> Interpreter -> Manager -> Controller -> Resource Manager -> Sensor Pool.

Q27 Explain the target-tracking case study in sensor-cloud.

Source basis: uploaded lecture PDF, pages 48 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A warehouse needs intrusion detection for six months. Buying and maintaining a dedicated target-tracking WSN is expensive, and an existing WSN owner will not transfer ownership or raw data. The warehouse only needs the tracking result as a service.

A. Concept [1 mark]: The case study compares costly dedicated WSN ownership with service-based target tracking in a sensor-cloud environment.

B. Scenario Evidence [1 mark]

- Dedicated network ownership is costly.
- The existing owner will not share the WSN directly.
- The organization needs application output, not physical sensors.

C. Theory Application [2.5 marks]

1. Traditional WSN deployment creates long-term setup and maintenance investment.
2. A refusal to share sensed information forces each organization to deploy its own network.
3. Sensor-cloud separates ownership from service consumption.
4. Virtualization and Sensor-as-a-Service let the warehouse use target tracking without owning the WSN.
5. Pay-per-use aligns cost with the six-month requirement.

D. Recommendation [1.5 marks]: Contract with an SCSP for a virtual target-tracking service, specifying coverage, QoS, privacy and six-month usage terms.

E. Expected Impact [1 mark]

- Avoided long-term capital investment.
- Faster deployment.
- Access to tracking while the sensor owner retains physical control.

Conclusion: Sensor-cloud makes target tracking more economical and accessible by separating application use from physical WSN ownership.

A+ EXAM SUGGESTION: Write: Problem -> Traditional consequence -> Sensor-cloud solution -> Benefit.

Q28 Summarize the historical cloud market analysis presented in the lecture.

Source basis: uploaded lecture PDF, pages 60-63 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A student is given lecture charts about cloud-market growth and leading providers from 2016-2017. The exam asks for analysis, but the student must not present the historical chart as current market truth.

A. Concept [1 mark]: The lecture provides a historical 2017 snapshot of cloud infrastructure growth and identifies the major provider companies of that period.

B. Scenario Evidence [1 mark]

- The data belongs to a dated lecture snapshot.
- The chart discusses growth and major providers.
- Interpretation must separate source content from present-day claims.

C. Theory Application [2.5 marks]

1. The lecture links the progression from mainframes to cloud infrastructure services.
2. It identifies business interest in SaaS and increasing infrastructure adoption.
3. It presents Amazon, Microsoft, IBM and Google as major participants in the historical snapshot.
4. Market percentages/rankings should be described as lecture-era evidence, not verified current statistics.

D. Recommendation [1.5 marks]: State the year and source context explicitly, summarize only the trend and named participants shown, and avoid updating or extrapolating beyond the lecture.

E. Expected Impact [1 mark]

- Source-faithful exam answer.
- No accidental use of outdated facts as current facts.
- Clear distinction between historical analysis and present market status.

Conclusion: The historical analysis demonstrates the commercialization and concentration of cloud services among major providers.

A+ EXAM SUGGESTION: Write the phrase: 'As presented in the lecture's 2017 market snapshot.'

Q29 Explain AWS and its major cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 63-67 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: An online media company needs resizable virtual servers, object storage, a virtual private network, a managed database and data-transfer services. The exam asks which AWS services from the lecture match these needs.

A. Concept [1 mark]: AWS is presented as a major global provider offering cloud products across computing, storage, databases, networking, IoT, management and security.

B. Scenario Evidence [1 mark]

- Compute must resize with demand.
- Large durable object storage is needed.
- Networking, database and transfer functions are also required.

C. Theory Application [2.5 marks]

1. Amazon EC2 supplies secure, resizable compute capacity.
2. Amazon S3 provides object storage through Simple Storage Service.
3. Virtual Private Cloud supports isolated networking.
4. The lecture also groups AWS services across databases, analytics, networking, mobile, IoT and enterprise tools.
5. AWS service breadth allows separate cloud components to form one solution.

D. Recommendation [1.5 marks]: Use EC2 for compute, S3 for objects, VPC for network isolation and the lecture-described managed database/transfer categories for supporting services; monitor usage-based cost.

E. Expected Impact [1 mark]

- Elastic media processing.
- Scalable storage.
- A modular cloud architecture instead of purchased infrastructure.

Conclusion: AWS combines elastic computing and managed storage with security, interfaces and development support.

A+ EXAM SUGGESTION: Write full forms: EC2 = Elastic Compute Cloud; S3 = Simple Storage Service.

Q30 Explain Microsoft cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 68-69 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A consulting firm wants subscription office applications for staff and a cloud environment for developers to build and host customer solutions. It must distinguish Office 365 from Microsoft Azure.

A. Concept [1 mark]: The lecture describes Microsoft's shift toward cloud technology through Office 365 and Microsoft Azure.

B. Scenario Evidence [1 mark]

- Staff need ready-to-use productivity software.
- Developers need compute, platform and deployment capabilities.
- The two needs belong to different service levels.

C. Theory Application [2.5 marks]

1. Office 365 is presented as a cloud-based subscription combining applications such as Excel and Outlook with online services.
2. It represents SaaS for end-user productivity.
3. Microsoft Azure is the broader cloud platform used for application development, hosting and cloud resources.
4. Azure can support platform and infrastructure functions depending on the service used.

D. Recommendation [1.5 marks]: Deploy Office 365 for standard employee productivity and Azure for custom applications; manage identity and access consistently across both.

E. Expected Impact [1 mark]

- Correct separation of SaaS and platform needs.
- Faster staff onboarding.
- Scalable application development and hosting.

Conclusion: Microsoft combines end-user SaaS with a broad cloud platform for application and hybrid-cloud deployment.

A+ EXAM SUGGESTION: Do not describe Office 365 and Azure as the same service model.

Q31 Explain IBM cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 70-71 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A regulated enterprise wants a private environment for sensitive data, public capacity for variable workloads and an object-storage/video service portfolio. The exam asks how IBM cloud services in the lecture address the case.

A. Concept [1 mark]: The lecture presents IBM as a provider of public, private and hybrid cloud services across IaaS, PaaS and SaaS.

B. Scenario Evidence [1 mark]

- Public, private and hybrid options are needed.
- Object storage is required.
- Private cloud and streaming video are named needs.

C. Theory Application [2.5 marks]

1. The lecture presents IBM support for public, private and hybrid cloud environments.
2. Its portfolio spans IaaS, PaaS and SaaS.
3. IBM Cloud Object Storage addresses data-object storage.
4. IBM Cloud Private addresses a controlled private-cloud environment.
5. IBM Cloud Streaming Video addresses managed video delivery.

D. Recommendation [1.5 marks]: Use a hybrid IBM design: Cloud Private for sensitive workloads, public services for elasticity, object storage for data and streaming video where required.

E. Expected Impact [1 mark]

- Workload placement by sensitivity.
- Elastic capacity without abandoning private control.
- Integrated storage and media services.

Conclusion: IBM's cloud portfolio emphasizes enterprise integration, controlled storage, private-cloud operation and managed services.

A+ EXAM SUGGESTION: Remember the three lecture names: Cloud Object Storage, Cloud Private, Cloud Streaming Video.

Q32 Explain Google Cloud services described in the lecture.

Source basis: uploaded lecture PDF, pages 72-74 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A data company needs virtual machines, an automatically scaling web platform, large-query analytics, a NoSQL database and a managed MySQL database. Map each requirement to Google Cloud services described in the lecture.

A. Concept [1 mark]: Google Cloud Platform provides compute, application hosting, storage, analytics, API and database services.

B. Scenario Evidence [1 mark]

- Both IaaS and PaaS are required.
- Analytics must process large queries.
- NoSQL and relational databases are both needed.

C. Theory Application [2.5 marks]

1. Google Compute Engine provides IaaS virtual machines and load balancing.
2. Google App Engine provides PaaS web-application hosting with automatic scaling.
3. BigQuery provides analytics for large datasets.
4. Datastore provides NoSQL data storage.
5. Cloud SQL provides managed MySQL database service.

D. Recommendation [1.5 marks]: Use Compute Engine only for workloads requiring VM control, App Engine for standard web services, BigQuery for analytics, Datastore for NoSQL and Cloud SQL for relational data.

E. Expected Impact [1 mark]

- Each workload uses the correct managed abstraction.
- Automatic scaling for web applications.
- Clear separation of analytics, NoSQL and SQL data services.

Conclusion: Google Cloud combines IaaS, PaaS, storage, data analytics and managed database services.

A+ EXAM SUGGESTION: Classify: Compute Engine-IaaS, App Engine-PaaS, BigQuery-analytics, Datastore-NoSQL, Cloud SQL-MySQL.

Q33 Explain Apple iCloud and its major features described in the lecture.

Source basis: uploaded lecture PDF, pages 75-78 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: An Apple-device user wants automatic backup, synchronized documents and photos, device location, password storage and access to a music library. Explain how the lecture's iCloud features meet the scenario.

A. Concept [1 mark]: iCloud is an Apple cloud-storage and cloud-computing service designed to synchronize, protect and access user information across devices.

B. Scenario Evidence [1 mark]

- Data must synchronize across devices.
- Backup and lost-device support are required.
- Files, photos, credentials and media must be managed.

C. Theory Application [2.5 marks]

1. iCloud synchronizes content across supported Apple devices and compatible systems.
2. iCloud Backup protects device data and Find My iPhone supports location/recovery.
3. iCloud Drive stores documents and app data.
4. Photo Stream/iCloud Photo Library handle photos and videos.
5. iCloud Keychain stores credentials, while iTunes Match supports music-library matching.

D. Recommendation [1.5 marks]: Enable backup, device location, Drive, photo synchronization and Keychain according to privacy needs, with strong account authentication.

E. Expected Impact [1 mark]

- Consistent cross-device access.
- Improved recovery after loss or failure.
- Central management of files, photos, passwords and media.

Conclusion: iCloud integrates storage, synchronization, backup, security, productivity and device-management services for Apple users.

A+ EXAM SUGGESTION: Group the answer under synchronization, backup/location, files/photos, security and media.

Q34 Define fog computing and explain its characteristics.

Source basis: uploaded lecture PDF, pages 81-85 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A factory has hundreds of IoT sensors. Sending every reading to a distant cloud causes delay and backbone traffic. Local gateways and routers can store, process and control data near the machines.

A. Concept [1 mark]: Fog computing moves storage, communication, control and computing functions from a distant cloud toward end-user or near- user edge devices.

B. Scenario Evidence [1 mark]

- Data is generated close to industrial equipment.
- Distant-cloud communication creates delay and traffic.
- Gateways/routers can perform local functions.

C. Theory Application [2.5 marks]

1. Fog computing moves compute, storage, communication and control toward end users/data sources.
2. Fog Computing Nodes can include routers, switches, access points, IoT gateways and set-top boxes.
3. Local storage and communication reduce constant backbone use.
4. Local management supports control, configuration and measurement.
5. A fog abstraction layer provides uniform allocation, monitoring, security, device management, storage and compute.

D. Recommendation [1.5 marks]: Deploy fog nodes at factory gateways for immediate filtering/control and send only summarized or archival data to the central cloud.

E. Expected Impact [1 mark]

- Lower response time.
- Reduced network traffic.
- Continued local operation with cloud support.

Conclusion: Fog computing reduces dependence on distant clouds by providing decentralized, heterogeneous and near-user services.

A+ EXAM SUGGESTION: Underline FCN and mention IP plus non-IP access support.

Q35 Explain Mobile Edge Computing (MEC), its characteristics and controller.

Source basis: uploaded lecture PDF, pages 86-90 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A mobile operator wants a real-time augmented-reality service. Processing in a distant cloud is too slow, and the application needs current radio-network load, user location and connected-device information.

A. Concept [1 mark]: MEC places cloud computing and IT services at the edge of a mobile cellular network.

B. Scenario Evidence [1 mark]

- The application is latency-sensitive.
- Cellular-network context is required.
- Processing can be placed near the base station.

C. Theory Application [2.5 marks]

1. MEC places cloud computing and IT services at the mobile cellular edge.
2. MEC servers can be co-located with an RNC or macro base station/eNB.
3. Near-edge processing reduces delay and congestion and improves reliability.
4. Authorized applications can access RAN capabilities and real-time context.
5. The Mobile Edge Orchestrator manages hosts, resources, topology, information flow and applications.

D. Recommendation [1.5 marks]: Deploy the AR application on MEC servers near selected base stations and use the MEO for placement, lifecycle and resource management.

E. Expected Impact [1 mark]

- Faster interactive response.
- Lower core-network traffic.
- Context-aware and more reliable service.

Conclusion: MEC combines cellular-network context with virtualized computing and storage near the base station.

A+ EXAM SUGGESTION: Write full forms: MEC, RAN, RNC, eNB/gNB and MEO.

Q36 Define cloudlet and explain its three-layer architecture.

Source basis: uploaded lecture PDF, pages 91-94 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A museum offers an interactive mobile vision application. Phones lack enough processing power, but a small Wi-Fi-connected compute box is available inside the museum and a distant cloud remains available for backup.

A. Concept [1 mark]: A cloudlet is a mobility-enhanced, small-scale cloud data center located near mobile devices at the edge of the Internet.

B. Scenario Evidence [1 mark]

- The application is resource-intensive and interactive.
- A nearby small data center is available.
- Low-latency Wi-Fi access is possible.

C. Theory Application [2.5 marks]

1. A cloudlet is a small, mobility-enhanced cloud data center near mobile devices.
2. It sits between user equipment and the distant cloud.
3. VM-based provisioning supplies resources in real time.
4. The Cloudlet layer is managed by a Cloudlet Agent.
5. The Node layer uses Node Agents for execution environments, and the Component layer provides services.

D. Recommendation [1.5 marks]: Run image processing in VMs on the museum cloudlet, use Wi-Fi for direct access and send only backup or large archival tasks to the central cloud.

E. Expected Impact [1 mark]

- Low-latency visitor interaction.
- Reduced phone computation and battery use.
- Local service with central-cloud fallback.

Conclusion: Cloudlets bring VM-based computing close to mobile users through a compact three-layer edge architecture.

A+ EXAM SUGGESTION: Draw UE <-> Cloudlet <-> Cloud and list Cloudlet, Node, Component layers.

Q37 Compare fog computing, MEC and cloudlet computing.

Source basis: uploaded lecture PDF, pages 94-95 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A smart-city designer has three projects: non-IP sensor gateways across neighborhoods, a cellular vehicle application needing radio context, and a campus mobile app needing a nearby Wi-Fi mini-data-center. Select fog, MEC or cloudlet for each.

A. Concept [1 mark]: Fog, MEC and cloudlet computing all move resources closer to users, but differ in node type, location, architecture, context awareness and access.

B. Scenario Evidence [1 mark]

- Project 1 needs heterogeneous gateways and non-IP support.
- Project 2 needs cellular context at a base station.
- Project 3 needs one-hop Wi-Fi compute near users.

C. Theory Application [2.5 marks]

1. Fog fits routers, gateways and access points distributed between devices and cloud, including non-IP support.
2. MEC fits servers at cellular base stations and provides high context awareness through the mobile network.
3. Cloudlet fits a data-center-in-a-box near users, usually reached by Wi-Fi with strong logical proximity.
4. Fog may be one or multiple hops; MEC and cloudlet are generally one hop.

D. Recommendation [1.5 marks]: Use fog for neighborhood sensors, MEC for connected vehicles and a cloudlet for the campus application; avoid treating the three as interchangeable.

E. Expected Impact [1 mark]

- Correct latency and access characteristics.
- Better context use.
- Efficient placement for each network environment.

Conclusion: The best approach depends on network type, context awareness, latency, access method and required proximity.

A+ EXAM SUGGESTION: Compare device, location, architecture, context, proximity, access and inter-node communication.

Q38 Explain the trends and issues that motivate mobile and IoT cloud computing.

Source basis: uploaded lecture PDF, pages 97-99 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A wearable-health application collects large datasets and must make time-sensitive decisions. The device has limited CPU, storage and battery, while users expect high data rates and QoS.

A. Concept [1 mark]: New mobile and IoT applications require high data rates, quality of service and processing capability beyond what many individual devices can supply.

B. Scenario Evidence [1 mark]

- Processing must finish within strict time limits.
- Data volume exceeds one device's capacity.
- Battery and local storage are limited.

C. Theory Application [2.5 marks]

1. Growing demand for high data rates and QoS motivates assistance beyond the device.
2. Advanced smartphones, sensors and actuators generate sophisticated workloads.
3. Limited CPU may not return results in time.
4. Situational awareness may require information from many nodes.
5. Large data volume and battery limits motivate cloud storage, computation offloading and edge processing.

D. Recommendation [1.5 marks]: Split the workload: process urgent health events at the edge, offload heavy analytics and long-term storage to the cloud, and keep minimal local processing for continuity.

E. Expected Impact [1 mark]

- Timely health decisions.
- Longer device battery life.
- Scalable analytics and storage.

Conclusion: These limitations motivate cloud and edge assistance for mobile and IoT computing.

A+ EXAM SUGGESTION: Connect each issue to a need: CPU->offloading, big data->storage, battery->remote compute, latency->edge.

Q39 Explain Mobile Cloud Computing (MCC) and its advantages.

Source basis: uploaded lecture PDF, pages 99-100 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A smartphone must execute a computation-heavy language-analysis task and store a large result archive. Local processing drains the battery, so the application can send work through LTE, the operator core and the Internet to a central cloud.

A. Concept [1 mark]: MCC allows mobile and IoT users to access computing and storage resources of a distant centralized cloud through the operator core network and Internet.

B. Scenario Evidence [1 mark]

- The device has limited computation and battery.
- A network path to a central cloud exists.
- Large storage is required.

C. Theory Application [2.5 marks]

1. MCC treats user equipment and IoT systems as cloud clients.
2. Requests travel through the mobile operator core network and Internet.
3. Complex or energy-consuming computation is offloaded to the central cloud.
4. Offloading can extend battery lifetime.
5. The cloud provides advanced SaaS/PaaS/IaaS functions and massive storage.

D. Recommendation [1.5 marks]: Use MCC for heavy analysis and archive storage, while controlling transfer size, authentication and fallback behavior when connectivity is weak.

E. Expected Impact [1 mark]

- Reduced local computation.
- Extended battery life.
- Access to powerful processing and large storage.

Conclusion: MCC expands mobile-device capability by borrowing centralized cloud computation, services and storage.

A+ EXAM SUGGESTION: Draw UE/IoT -> LTE/Core Network -> Internet -> Central Cloud.

Q40 Compare conventional MCC and edge computing.

Source basis: uploaded lecture PDF, pages 101-102 | Pattern: A-E matched answer | 7 marks

BIG SCENARIO: A live mobile video service currently sends all processing to a distant central cloud. Users experience long delay, jitter and high battery/network use. The operator can deploy smaller compute resources near base stations, but those edge resources have limited capacity.

A. Concept [1 mark]: Conventional MCC uses distant centralized clouds, while edge computing moves computing and storage closer to the user equipment.

B. Scenario Evidence [1 mark]

- The current centralized path causes latency and jitter.
- Real-time multimedia needs high QoS.
- Edge resources are close but less powerful.

C. Theory Application [2.5 marks]

1. Conventional MCC is centralized, far from the user and has long latency/high jitter.
2. It offers abundant compute and storage but uses more network resources and device energy.
3. Edge computing is distributed near base stations or access points, producing short delay and low jitter.
4. Edge reduces wide-area traffic and battery use but has limited compute/storage.
5. Prediction, pre-fetching and distributed control add complexity at the edge.

D. Recommendation [1.5 marks]: Process real-time video functions at the edge and retain the central cloud for large storage, global analytics and overflow; orchestrate workload movement according to capacity.

E. Expected Impact [1 mark]

- Lower delay and jitter.
- Better real-time QoS with reduced network/battery use.
- Central-cloud scale retained for heavy and archival tasks.

Conclusion: Edge computing improves latency, jitter and efficiency, whereas MCC provides greater centralized resource capacity.

A+ EXAM SUGGESTION: Use a comparison table: deployment, distance, latency, jitter, compute and storage.